Aliaa Mohamed Abdo

AI / ML Engineer

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Summary

AI/ML Engineer and Computer Science student at Helwan University (GPA 3.41 out of 4.00) specializing in **Computer Vision, Deep Learning, and NLP**. Proven track record building end-to-end intelligent pipelines using **Genetic Algorithms, PSO, LayoutLM, and FaceNet**, with hands-on deployment via **Docker, FastAPI, and Streamlit**. Passionate about turning complex data into measurable, production-ready solutions.

Education

Helwan University, Faculty of Computers and Artificial Intelligence — Sep 2023 – Jul 2027 (Expected)
B.Sc. in Artificial Intelligence (Senior Student)

- Cumulative **GPA: 3.41 out of 4.00** – Overall academic evaluation: Excellent.
- Relevant Coursework: Deep Learning, Neural Networks, Computer Vision, NLP, Metaheuristics & Evolutionary Algorithms, Optimization, Data Structures & Algorithm Design.

Experience

Freelance AI/ML Developer — 2024 – Present
Independent Research & Client Projects

- Engineered and evaluated **6 production-grade AI models** (LayoutLM, FaceNet, MobileNetV2), achieving **95 percent prediction accuracy** and reducing manual review effort by an estimated **40 percent**.
- Applied metaheuristic frameworks (**Genetic Algorithms & PSO**) to optimize feature subsets, reducing input dimensionality by **73 percent** and cutting model training time by **35 percent**.
- Built high-performance OpenCV preprocessing pipelines that reduced feature extraction latency by **20 percent**, enabling real-time inference on resource-constrained environments.
- Cut containerized deployment overhead by **30 percent** via modular **Docker** and **FastAPI** microservice architecture, reducing average release cycle from days to hours.
- Maintained collaborative **GitHub** workflows across multi-member teams, enforcing branch strategies that ensured **100 percent reproducibility** across all experiment runs.

Skills

AI & Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Multi-Objective Optimization, Evolutionary Computation (GA, PSO), Model Evaluation, Statistical Analysis & EDA.

Deep Learning, CV & NLP: CNNs, MobileNetV2, FaceNet, Transformers, BERT Embeddings, LayoutLM (HuggingFace), Face Detection (OpenCV), OCR (Tesseract).

Frameworks & Libraries: PyTorch, TorchVision, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Plotly.

Languages & Tools: Python, Docker, FastAPI, Streamlit, Git & GitHub, SQL, Java.

Soft Skills: Problem-solving, Team Collaboration, Time Management, Critical Thinking, Active Communication.

Professional Training & Internships

Digital Egypt Pioneers Initiative (DEPI) — Aug 2026 – Present
Machine Learning Specialist Trainee (Round 5)

- Enrolled in an intensive professional track covering advanced **Machine Learning, Deep Learning, Computer Vision (CV), and Natural Language Processing (NLP)**.
- Developing end-to-end applied AI pipelines, from data engineering and model architecture design to real-world deployment and evaluation.
- Collaborating on production-focused computer vision and NLP capstone projects adhering to industry-standard workflows.

Sprints — Jan 2025 – Feb 2025
Machine Learning Engineering Track

- Designed and executed end-to-end ML pipelines for **3 industry-level production scenarios**, covering full EDA, feature engineering, and model deployment phases.
- Improved model F1-scores by **15 percent** through systematic hyperparameter tuning using Scikit-learn cross-validation strategies.
- Processed and partitioned over **1,000 data samples** with strict train/test isolation to eliminate data leakage and ensure generalizable model performance.

IEEE Student Branches Workshops Feb 2025 – Sep 2025
Software Development & UI Integration Track

- Developed responsive web applications using **JavaScript (ES6+), React.js, and Tailwind CSS**, following component-driven UI/UX design patterns.
- Collaborated cross-functionally within a team environment to integrate real-time **API endpoints** with reactive frontend state management.
- Bridged data-driven ML backends with interactive user interfaces, delivering end-to-end full-stack prototypes within sprint deadlines.

Projects

Invoice Document Intelligence System [GitHub] 05/2026

- Engineered a hybrid Document AI pipeline fine-tuning **LayoutLM** (HuggingFace) and Tesseract OCR, reaching a peak **F1-score of 0.9646** on multi-field invoice extraction.
- Built a production-ready **FastAPI** microservice returning structured JSON data and field-level confidence scores with rule-based fallback verification.
- Integrated **AWS S3** file storage and containerized the end-to-end service using **Docker Compose** alongside an interactive dashboard.

Containerized Face Mask Detection Pipeline [GitHub] 04/2026

- Developed an end-to-end deep learning classifier using **MobileNetV2** transfer learning, trained and evaluated across **12,000 real-world face images**.
- Implemented an automated **OpenCV Haar Cascade** face detection and cropping pipeline (with a 20 percent margin expansion) to ensure high inference accuracy on full-body photos.
- Packaged the service into a containerized **Docker** environment with a **FastAPI** REST endpoint and interactive web UI.

Face Recognition Attendance System [GitHub] 05/2026

- Architected a modular biometric pipeline extracting **FaceNet512 embeddings** and classifying identities with an RBF-kernel **SVM**, achieving **98.50 percent 5-Fold CV accuracy**.
- Integrated CLAHE illumination normalization and Haar Cascade face alignment, boosting baseline verification accuracy by **15 percent**.
- Automated real-time camera inference and daily CSV attendance logging with duplicate verification and an interactive web visualizer.

GA-Based Feature Selection for TB Classification [GitHub] 11/2025

- Built a metaheuristic optimization framework using **Genetic Algorithms** and Binary PSO, reducing input dimensionality by **73 percent** while preserving classification signal.
- Achieved **93.57 percent classification accuracy** across 30 independent trials, isolating the top **5 percent** discriminative features via 3-Fold KNN.
- Extracted a 36-dimensional handcrafted feature vector using OpenCV and deployed an interactive **Streamlit** analytics dashboard.

Certifications

- **Machine Learning Engineering** – TechVerse Bootcamp, IEEE FCI-Helwan (Sep 2025).
- **Machine Learning Hackathon** – Helwan Cyber Arena (Sep 2025).
- **Machine Learning Engineering Track** – Sprints (Feb 2025).
- **Artificial Intelligence Fundamentals** – Huawei AI Academy.

Languages

- **Arabic:** Native speaker.
- **English:** Professional Working Proficiency.